

Cs601 short notes

Composed by Amir

Midterm preparation

Topics 1 to 113

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Define Data communication?

Data Communication is the exchange of data (in the form of 0's and 1's) between two devices (computers) via some form of the transmission medium. Data communication is communication between computers.

Describe Components of a Data Communication system.

A data communications system has five components.



- Message
- Sender
- Receiver
- Medium
- Protocol

Difference between LOCAL and REMOTE Data Communication

LOCAL Data communication

- Sender
- Receiver
- Medium
- Protocol

Difference between LOCAL and REMOTE Data Communication

LOCAL Data communication

is considered to be local if the communicating devices are present in the same building or a similarly restricted geographical area.

REMOTE Data

Communication is considered remote, if the devices are **farther apart**.

Effectiveness of a Data Communication System:

Effectiveness of a data communications system depends on which four characteristics?

The effectiveness of a data communications system depends on four fundamental characteristics: **delivery, accuracy, timeliness, and jitter**.

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Describe Message

A message (verbal or nonverbal, or both) is the content of the communication process. The originator of the message in the communication process is the sender. In short anything that can be represented using binary bits.

SENDER

A source or sender is one of the basic concepts of communication and

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SENDER

A source or sender is one of the basic concepts of communication and information processing.

Sources are objects which encode message data and transmit the information, via a channel, to one or more observers.

RECEIVER

In the communication process, the receiver is the **listener, reader, or observer**—that is, the individual (or the group of individuals) to whom a message is directed. The receiver is also called the "audience" or decoder.

Medium

In the communication process, a medium is a **channel or system of communication**—

the means by which information (the message) is transmitted between a speaker or writer (the sender) and an audience (the receiver). The plural form is media, and the term is also known as a channel.

A communication protocol is a **system of rules that allows two or more entities of a communications system to transmit information via any kind of variation** of a

physical quantity. The protocol defines the rules, syntax, semantics and synchronization of communication and possible error recovery methods.

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Describe KEY DATA COMMUNICATION TERMINOLOGY

- **Link:** connects adjacent nodes Wires, Cables, Anything that physically connects two nodes TM Path: end-to-end route within a network TM
- **Circuit:** the conduit over which data travels TM
- **Packetizing:** dividing messages into fixed-length packets prior to transmission over a network's communication media TM
- **Routing:** determining a message's path from sending to receiving nodes The transmission medium may itself be a network, so route needs to be specified

Types of Transmission Media

Types of Transmission Media

<u>Medium</u>	<u>Speed</u>	<u>Cost</u>
<u>Twisted Wire</u>	<u>300bps-10Mbps</u> www.thundershare.net	<u>Low</u>
<u>Microwave</u>	<u>256Kbps-100Mbps</u>	<u>Low</u>
<u>Coaxial Cable</u>	<u>56Kbps-200Mbps</u>	<u>Low</u>
<u>Fiber Optic Cable</u>	<u>500Kbps-10Gbps</u>	<u>High</u>

What is Network?how many types of networks?

“A NETWORK is a set of devices (Nodes) connected by Communication Links”

- Node: Can be a Computer, Printer or any other device capable of sending or receiving.

Three Main Classifications of the Networks

- LANS www.thundershare.net
- MANS
- WANS

How could it be more secure using distributed processing?

In this case security is achieved by providing the user with limited access. It means that user can only perform those tasks which are allowed by system designer. Its' simple example is bank's ATM. Where user can't perform operations on database other than provided

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Describe Network Criteria

A network must be able to meet a certain number of criteria:

- Performance
 - Throughput
 - Delay

- Reliability

- Security

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Performance

Can be measured in many ways including Transit and Response

Time Depends on a no. of Factors:

- Number of USERS
- Type of Transmission Medium
- Hardware
- Software
- Network Criteria

• Network Criteria

Reliability

Depends on a no. of Factors:

- Frequency of Failure

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- Recovery Time of a Network after Failure
- Catastrophe
- Fire, Earthquake or Theft

Security

Unauthorized Access

Security

Unauthorized Access

- Sensitive data
- Protection at multiple levels:
- Lower level: Passwords and user ID codes
- Upper Level: Encryption

What is a sample?

Sample is the amplitude of an analog signal at some specific interval. It is achieved in pulse amplitude modulation, which is the first step towards analog to digital conversion.

What is Protocol Layering

Simple Communication: only one simple protocol
Complex Communication, we need a protocol at each layer, or Protocol Layering.

Advantages:

- Modularity
- Separation of Service & Implementation

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Advantages:

- Modularity
- Separation of Service & Implementation
- Reduced Complexity & Cost

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Disadvantages

None Really!

How many Principles of Protocol Layering ?

Two Principles

- Bidirectional Communication

Disadvantages

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How many Principles of Protocol Layering ?

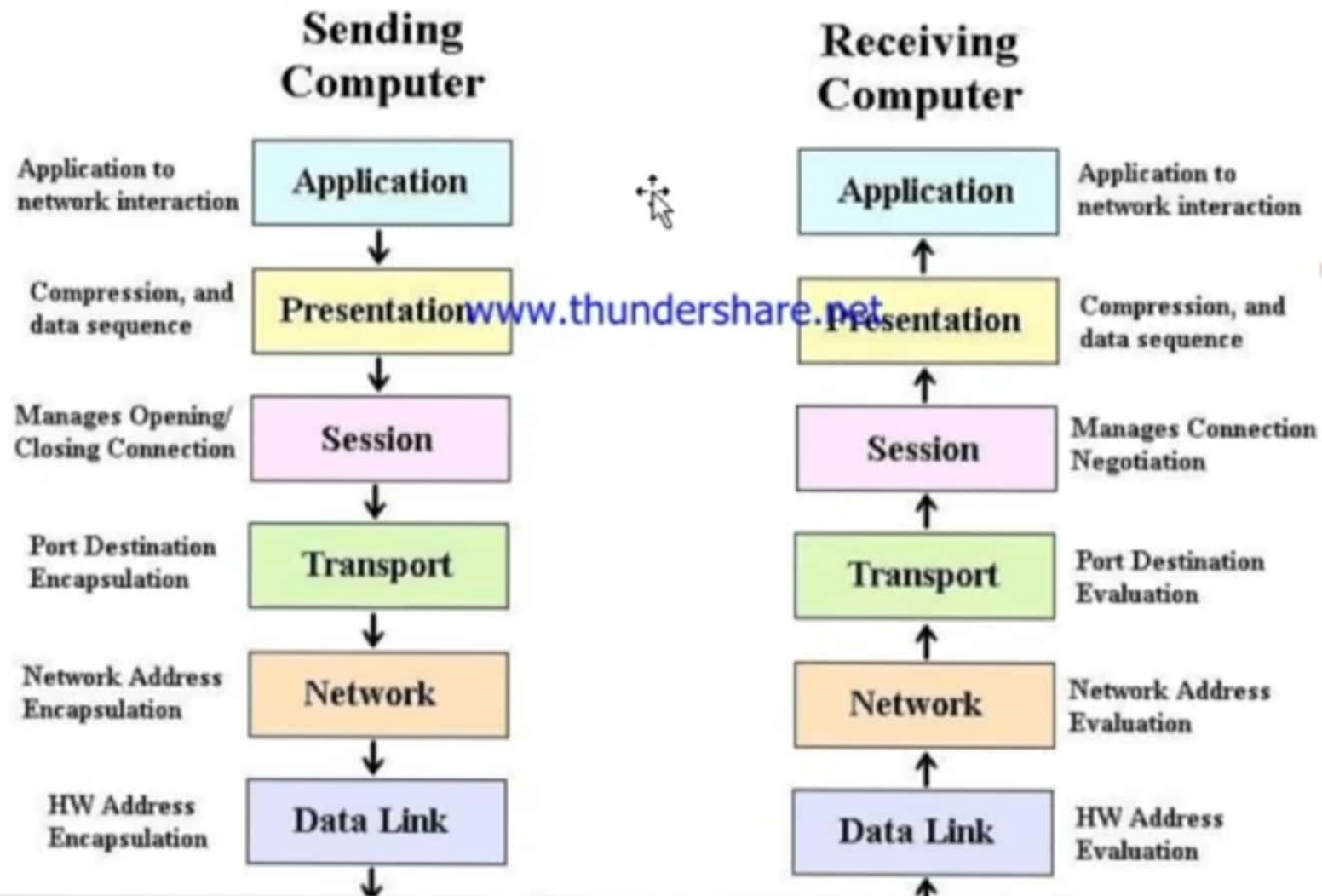
Two Principles

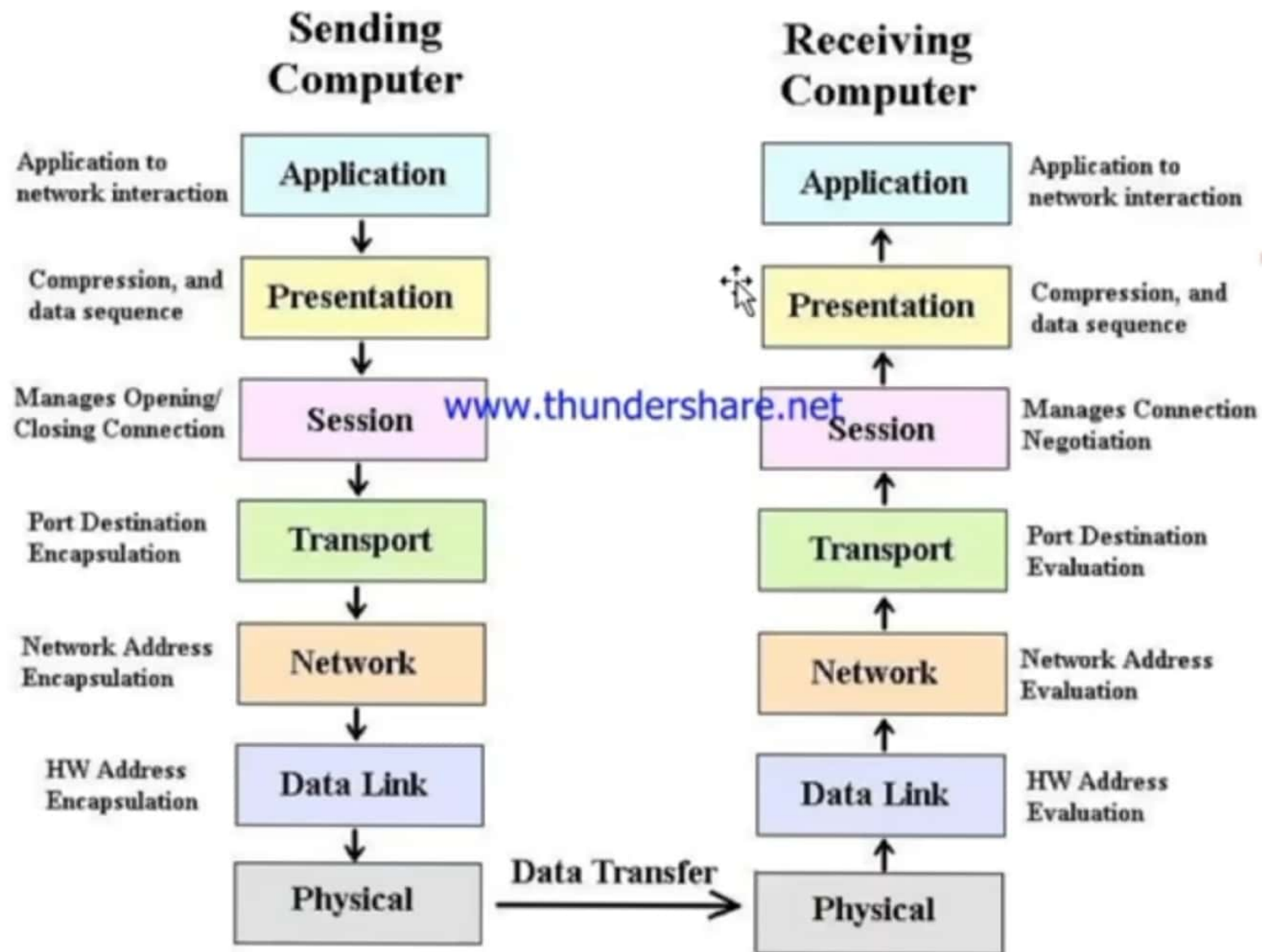
- Bidirectional Communication
- Two objects under each layer at both sites should be identical

TCP/IP Protocol Suite

- Protocol suite used in Internet today
- Each Layer provides specific functionality
- Hierarchical Protocol
- Presented in 1973 and chosen to be the official protocol of Internet in 1983

Network Layer Interaction







What do we mean by the term session?

Session is the period of time a user interfaces with an application. The user session begins when the user accesses the application and ends when the user quits the application.

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What is transit and response time mean?

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Transit time is the time taken by the message to travel from source to destination. Response time is the time of response (acknowledgement) from destination to source.

What do you know about peak load periods

Peak load periods are those periods of time in which number of users on a network are more than normal time.

What are the advantages of networking?

Peak load periods are those periods of time in which number of users on a network are more than normal time.



What are the advantages of networking?

- Files can be stored on a central computer (the file server) allowing data to be shared throughout an organization.
- Files can be backed up more easily when they are all on a central files server rather than when they are scattered across a number of independent workstations.
- Networks also allow security to be established, ensuring that the network users may only have access to certain files and applications.
- Software and resources can be centrally managed.
- Network versions of software often allow for their speedy installation on workstations from the file server.
- Expensive devices such as laser printers or scanners can be shared.
- Users can access their files from any workstation

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What is the definition of Hub?

Hub is a common connection point for devices in a network. Hubs are commonly used to connect segments of a LAN. A hub contains multiple ports.

What is the difference between radio and microwave?



Although there is no clear-cut demarcation between radio and microwave, electromagnetic waves ranging in frequencies between 3 KHz and 1 GHz are normally called radio waves; waves ranging in frequencies between 1 GHz and 300 GHz are called microwaves. Radio waves are used for multicast communications, such as radio and television systems.

What is the difference between hub and switch?

On the outside, hubs and switches appear very similar in that they both have a number of RJ-45 jacks for connecting devices. Inside, however, they work very differently. To understand why switches provide so much more functionality than hubs, you must understand a fundamental limitation of (non-switched) Ethernet:

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How many layers of OSI model?

The OSI has 7 layers :

- Application layer
- Presentation layer
- Session layer
- Transport layer
- Network layer
- Data link layer

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- Physical layer

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What are routers?

Router: Routers relay packets among multiple interconnection networks. Router is device that determines the next network point to which a data packet should be forwarded enroute toward its destination. Routers create or maintain a table of the available routes and use this information to determine the best route for a given data packet.



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What do we mean by capacity of the channel?

Channel capacity: The maximum possible information transfer rate through a channel, subject to specified constraints.

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What is remote data communication?

Data Communication is considered remote, if the devices are farther apart.

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Define encryption of data. www.thundershare.net

Encryption of data: The translation of data into a secret code. Encryption is the most effective way to achieve data security. To read an encrypted file, you must have access to a secret key or password that enables you to decrypt it. Unencrypted data is called plain text; encrypted data is referred to as cipher text.

What is distributed processing?

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What is attenuation?

Attenuation is signal loss due to the diminishing availability of signal energy, or signal power. High attenuation may lead to the inability to recover the signal on the far end. Signal repeaters may be used on the transmission path to periodically boost the signal strength. Baseband transmission is extremely limited to attenuation. Broad-band much less so, In addition, wireless communications is much less susceptible to attenuation than is wire-line communications such as x-



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Regards M.Amir

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